

Demonstration of Proposed Features

Date	24 June 2026
Team ID	SWTD-2026-3249
Project Name	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed OptiCrop solution.

S. No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	ML-Based Crop Recommendation Engine	Predicts optimal crop choice and expected yield using soil, weather, and historical data.	Implemented	Yes	Core model trained and validated on Kaggle/UCI-style datasets.
2	Real-Time Irrigation & Fertilizer Advisory	Generates weather-aware, field-specific irrigation and fertilizer dosage recommendations.	Implemented	Yes	Live weather data integrated via OpenWeatherMap API.
3	Farmer Mobile App (Offline-First)	Flutter-based mobile app allowing farmers to view recommendations with limited connectivity.	Implemented	Yes	Core screens and offline caching demonstrated on Android.
4	SMS Alerts for Non-Smartphone Users	Sends key advisories via SMS using Twilio for farmers without smartphone access.	Partial	Yes	Basic alert flow working; regional language templates still in progress.
5	Geospatial Field Mapping	Lets farmers mark field boundaries and view location-based insights via Google Maps API.	Implemented	Yes	Field boundary drawing and storage demonstrated end-to-end.
6	Continuous Learning from Farmer Feedback	Refines recommendation models each season using farmer-reported outcomes.	Partial	No	Feedback capture UI built; retraining pipeline scheduled for next iteration.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	4 fully implemented, 2 partially implemented
Total Features Demonstrated	5 of 6
Overall Implementation Rate (%)	~83% (counting partial features as half-complete)